

JEE MAIN REVISION MASTER: ALDEHYDE AND KETONE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 22 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Oxidation, Reduction & Identification Tests	12 PYQs	Tests & Reductions
High	Nucleophilic Addition & Carbonyl Condensation	10 PYQs	Carbonyl Reactions

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Oxidation, Reduction & Identification Tests	$\text{Iodoform test is positive for } \text{CH}_3\text{CO-} \text{ or } \text{CH}_3\text{CH(OH)- groups} \implies \text{CHI}_3 \text{ (yellow ppt)}$
Nucleophilic Addition & Carbonyl Condensation	$\text{Aldol: Requires } \alpha\text{-H} \rightarrow \alpha, \beta\text{-unsaturated carbonyl}$, $\text{Cannizzaro: No } \alpha\text{-H} \rightarrow \text{Alcohol} + \text{Carboxylate salt}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Oxidation, Reduction & Identification Tests (High Yield — 12 PYQs)

Core Principle: Silver mirror Tollens' test, Fehling's cuprous reduction, yellow iodoform precipitation, Clemmensen (Zn-Hg/HCl), and Wolff-Kishner ($\text{NH}_2\text{NH}_2/\text{KOH}$) carbonyl deoxygenation.

■ **Caution / JEE Pitfall:** Benzaldehyde reduces Tollens' reagent to give silver mirror, but does NOT reduce Fehling's solution!

■ **Speed Trick:** Clemmensen reduction is suitable for acid-stable compounds. For acid-sensitive carbonyls, use Wolff-Kishner reduction (basic medium).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The compound that does not answer iodoform test is..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "Oxidation of Toluene with chromyl chloride followed by hydrolysis gives Benzaldehyde. This reaction is known as _____" [Ans: N/A]
- [2022 Shift] Q#1 (MCQ): "The test to differentiate between pentan-2-one and pentan-3-one is..." [Ans: N/A]

3.2 Nucleophilic Addition & Carbonyl Condensation (High Yield — 10 PYQs)

Core Principle: Reversible nucleophilic additions, Aldol condensation mechanism, and Cannizzaro disproportionation of non-enolizable aldehydes.

■ **Caution / JEE Pitfall:** Benzaldehyde and Formaldehyde lack α -hydrogen atoms, so they undergo CANNIZZARO reaction with concentrated NaOH, NOT Aldol condensation!

■ **Speed Trick:** Reactivity toward nucleophilic addition: $\text{HCHO} > \text{CH}_3\text{CHO} > \text{CH}_3\text{COCH}_3$ (decreases with steric crowding and +I electron donation).

Representative Source PYQs:

- **[2026 Shift] Q#2 (MCQ):** "The major product 'A' in the given reaction is Benzaldehyde $\rightarrow$ $\leq$ $\rightarrow$ 'A' (Major product)..." **[Ans: N/A]**
- **[2026 Shift] Q#3 (MCQ):** "Match the reagents in List-I with products obtained from their carbonyl compounds in List-II. List-I List-II (a) NH_2OH (i..." **[Ans: N/A]**
- **[2025 Shift] Q#1 (MCQ):** "Arrange the following compounds in their decreasing order of reactivity towards Nucleophilic addition reaction. $\text{CH}_3\text{COCH}_3, \text{CH}_3\text{COC}_2\text{H}_5$..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The compound that does not answer iodoform test is

- (A) Ethanal
- (B) Acetone
- (C) Ethanoic acid
- (D) Acetophenone

Correct Answer: None

Solution Analysis: The compound that does not answer the iodoform test is: Option C: Ethanoic acid Iodoform test is given by compounds containing the group $-\text{COCH}_3$. This is called the methyl ketone group. Some alcohols which can be oxidised to methyl ketones also give this test. Now check the options: Ethanal has the structure: CH_3CHO . It gives iodoform test. Acetone has the structure: CH_3COCH_3 . It contains the methyl ketone group, so it gives iodoform test. Acetophenone has the structure: $\text{C}_6\text{H}_5\text{COCH}_3$. It also contains the methyl ketone group, so it gives iodoform test. Ethanoic acid

Benchmark Problem 2: KCET 2026 (Q#2)

The major product 'A' in the given reaction is Benzaldehyde $\rightarrow$ $\leq$ $\rightarrow$ 'A' (Major product)

- (A)
- (B)
- (C)
- (D)

Correct Answer: None

Solution Analysis: This reaction is a crossed aldol condensation. Since it occurs between benzaldehyde and acetophenone in the presence of a base, it is also called Claisen-Schmidt condensation. Benzaldehyde ($\text{C}_6\text{H}_5\text{CHO}$) does not have any α -hydrogen. So, it cannot form an enolate ion. Acetophenone ($\text{C}_6\text{H}_5\text{COCH}_3$) has α -hydrogens, so the base removes one α -hydrogen from it. This gives the enolate ion of acetophenone:

$$\text{C}_6\text{H}_5\text{COCH}_2^- + \text{OH}^- \rightleftharpoons \text{C}_6\text{H}_5\text{COCH}_2 + \text{OH}^-$$